WEEK 7: SQL Query Implementation for E-Commerce Database

-ATCHAYA S[E25AI106]

Scenario:

The e-commerce company has collected customer, product, order, and inventory data. In this week, SQL queries are implemented to search, filter, sort, and generate reports from the existing Phone Case E-Commerce Database. The queries help retrieve customer and product information, identify product availability, analyze orders and payments, and generate basic business reports.

1. Perform basic SQL queries

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMER;

CUSTO CUSTOMER_NAME                  EMAIL                          PHONE
----- ------------------------------ ------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
C001  Arun Kumar                     arun@gmail.com                 9876543210
Chennai
10-SEP-26

C002  Aakash                         aakash35@gmail.com             9876543211
Tambaram
11-SEP-26

CUSTO CUSTOMER_NAME                  EMAIL                          PHONE
----- ------------------------------ ------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

C003  Rahul Raj                      rahul@gmail.com                9876543212
Chengalpattu
12-SEP-26

C004  Divya Kumar                    divya@gmail.com                9876543213
Velachery

CUSTO CUSTOMER_NAME                  EMAIL                          PHONE
----- ------------------------------ ------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
13-SEP-26

C005  Karthik S                      karthik@gmail.com              9876543214
Guindy
```

```
C006  Priya Sharma                    priya@gmail.com                9876543210

CUSTO CUSTOMER_NAME                   EMAIL                          PHONE
----- ------------------------------  ------------------------------ ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
Chennai
10-OCT-26


6 rows selected.
```

2. Display all available products.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000001
Clear Shock Case
CAT0001        299              50

P000002
Soft Silicone Case
CAT0002        249              45

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P000003
Crystal Clear Case
CAT0003        199              60

P000004
Floral Printed Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
CAT0004        349              35
```

```
P000005
Luxury Designer Case
CAT0005       599            25

P000006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Personal Photo Case
CAT0006       449            40

P000007
Heavy Duty Case
CAT0007       499            30


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000008
Matte Basic Case
CAT0001       199            40

P000009
Classic Black Case
CAT0001       229            35
```

```
P000010
Slim Basic Case
CAT0001        249             30

P000011
Blue Silicone Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
CAT0002        279             45

P000012
Pink Silicone Case
CAT0002        279             40

P000013

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Soft Grip Case
CAT0002        299             35

P000014
Clear Glitter Case
CAT0003        249             50
```

```
PRODUCT
-------
PRODUCT_NAME
-------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000015
Crystal Edge Case
CAT0003        269             45

P000016
Ultra Clear Case
CAT0003        229             55

PRODUCT
-------
PRODUCT_NAME
-------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P000017
Floral Print Case
CAT0004        349             30

P000018
Cartoon Print Case

PRODUCT
-------
PRODUCT_NAME
-------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
CAT0004        329             35

P000019
Abstract Art Case
CAT0004        379             25
```

```
PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Rugged Protection Case
CAT0007        549             20

P000028
Military Grade Case
CAT0007        599             15


PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000029
Everyday Basic Case
CAT0001        219             30

P000030
Simple White Case
CAT0001        239             25

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------


30 rows selected.
```

3. Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_NAME
------------------------------------------------------------
     PRICE
----------
Clear Shock Case
       299

Soft Silicone Case
       249

Crystal Clear Case
       199


PRODUCT_NAME
------------------------------------------------------------
     PRICE
----------
Floral Printed Case
       349

Luxury Designer Case
       599

Personal Photo Case
       449


PRODUCT_NAME
------------------------------------------------------------
     PRICE
----------
Heavy Duty Case
       499

Matte Basic Case
       199
```

```
Classic Black Case
       229


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
Slim Basic Case
       249

Blue Silicone Case
       279

Pink Silicone Case
       279


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
Soft Grip Case
       299

Clear Glitter Case
       249

Crystal Edge Case
       269


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
Ultra Clear Case
       229

Floral Print Case
       349
```

```
Cartoon Print Case
       329


PRODUCT_NAME
----------------------------------------------------
     PRICE
----------
Abstract Art Case
       379

Elegant Designer Case
       549

Premium Pattern Case
       649


PRODUCT_NAME
----------------------------------------------------
     PRICE
----------
Luxury Gold Case
       699

Family Photo Case
       449

Collage Photo Case
       499


PRODUCT_NAME
----------------------------------------------------
     PRICE
----------
Custom Memory Case
       479
```

```
Shockproof Armor Case
       499

Rugged Protection Case
       549


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Military Grade Case
       599

Everyday Basic Case
       219

Simple White Case
       239


30 rows selected.
```

4. Display all orders placed by customers.

```
SQL> SELECT *
  2  FROM ORDERS;

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ --------------------
      1001 C001  20-SEP-26         1299 Pending
      1002 C002  21-SEP-26          899 Shipped
      1003 C003  22-SEP-26         1599 Delivered
      1004 C001  23-SEP-26          699 Pending
      1005 C004  24-SEP-26          450 Shipped
      1006 C005  24-SEP-26          750 Pending
      1007 C002  25-SEP-26         1200 Delivered
      1008 C003  25-SEP-26          550 Shipped
      1009 C005  26-SEP-26          999 Pending
      1012 C005  24-SEP-26          550 Pending
      1011 C004  24-SEP-26          500 Delivered

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ --------------------
      1010 C004  24-SEP-26          550 Shipped
      1017 C001  27-SEP-26          449 Delivered
      1013 C002  28-SEP-26          499 Delivered
      1014 C003  29-SEP-26          199 Delivered
      1015 C004  30-SEP-26          229 Delivered
      1016 C005  01-OCT-26          249 Delivered

17 rows selected.
```

5. Retrieve payment details.

```
SQL> SELECT * FROM Payment;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         1       1001 20-SEP-26 UPI                  Completed
              1299

         2       1002 21-SEP-26 Credit Card          Completed
               899

         3       1003 22-SEP-26 Debit Card           Completed
              1599


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         4       1004 23-SEP-26 UPI                  Pending
               699

         5       1005 24-SEP-26 Credit Card          Completed
               450

         6       1006 24-SEP-26 Cash                 Pending
               750


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         7       1007 24-SEP-26 UPI                  Completed
              1200

         8       1008 25-SEP-26 Debit Card           Completed
               550
```

```
PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
        11       1011 24-SEP-26 UPI                  Completed
               450

        14       1014 29-SEP-26 Debit Card           Completed
               199

        10       1010 24-SEP-26 Credit Card          Failed
               550


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
        12       1012 24-SEP-26 Credit Card          Completed
               550

        13       1013 28-SEP-26 UPI                  Completed
               499

        15       1015 30-SEP-26 Credit Card          Completed
               229


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
        16       1016 01-OCT-26 UPI                  Completed
               249

        17       1017 27-SEP-26 Cash                 Pending
               449


17 rows selected.
```

2. Apply filtering conditions using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE > 5000;

no rows selected
```

2. Display products available in stock.

```
PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
STOCK_QUANTITY
--------------
Rugged Protection Case
           20

P000028
Military Grade Case
           15


PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
STOCK_QUANTITY
--------------
P000029
Everyday Basic Case
           30

P000030
Simple White Case
           25

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
STOCK_QUANTITY
--------------


30 rows selected.
```

3. Display completed orders.

```
SQL> SELECT *
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'DELIVERED';

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ --------------------
      1003 C003  22-SEP-26         1599 Delivered
      1007 C002  25-SEP-26         1200 Delivered
      1011 C004  24-SEP-26          500 Delivered
      1017 C001  27-SEP-26          449 Delivered
      1013 C002  28-SEP-26          499 Delivered
      1014 C003  29-SEP-26          199 Delivered
      1015 C004  30-SEP-26          229 Delivered
      1016 C005  01-OCT-26          249 Delivered

8 rows selected.
```

4. Find products with rating above 4.

```
PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
P000025
Custom Memory Case

P000001
Clear Shock Case

P000003
Crystal Clear Case


PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
P000019
Abstract Art Case

P000026
Shockproof Armor Case

P000028
Military Grade Case


PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
P000023
Family Photo Case


19 rows selected.
```

3. Sort data using ORDER BY

1. Display products from lowest to highest price.

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
P000026
Shockproof Armor Case
       499

P000024
Collage Photo Case
       499

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------

P000007
Heavy Duty Case
       499

P000027
Rugged Protection Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       549

P000020
Elegant Designer Case
       549
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
Military Grade Case
       599

P000005
Luxury Designer Case
       599


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
P000021
Premium Pattern Case
       649

P000022
Luxury Gold Case
       699

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------


30 rows selected.
```

2. Display customers alphabetically.

```
SQL> SELECT CUSTOMER_ID, CUSTOMER_NAME
  2  FROM CUSTOMER
  3  ORDER BY CUSTOMER_NAME ASC;

CUSTO CUSTOMER_NAME
----- ------------------------------
C002  Aakash
C001  Arun Kumar
C004  Divya Kumar
C005  Karthik S
C006  Priya Sharma
C003  Rahul Raj

6 rows selected.
```

3. Find the top 5 most expensive products.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM (
  3      SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  4      FROM PRODUCT
  5      ORDER BY PRICE DESC
  6  )
  7  WHERE ROWNUM <= 5;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P000022
Luxury Gold Case
       699

P000021
Premium Pattern Case
       649

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

P000005
Luxury Designer Case
       599

P000028
Military Grade Case
```

4. Display latest orders first.

```
SQL> SELECT *
  2  FROM ORDERS
  3  ORDER BY ORDER_DATE DESC;

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ --------------------
      1016 C005  01-OCT-26          249 Delivered
      1015 C004  30-SEP-26          229 Delivered
      1014 C003  29-SEP-26          199 Delivered
      1013 C002  28-SEP-26          499 Delivered
      1017 C001  27-SEP-26          449 Delivered
      1009 C005  26-SEP-26          999 Pending
      1008 C003  25-SEP-26          550 Shipped
      1007 C002  25-SEP-26         1200 Delivered
      1011 C004  24-SEP-26          500 Delivered
      1006 C005  24-SEP-26          750 Pending
      1012 C005  24-SEP-26          550 Pending

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ --------------------
      1005 C004  24-SEP-26          450 Shipped
      1010 C004  24-SEP-26          550 Shipped
      1004 C001  23-SEP-26          699 Pending
      1003 C003  22-SEP-26         1599 Delivered
      1002 C002  21-SEP-26          899 Shipped
      1001 C001  20-SEP-26         1299 Pending

17 rows selected.
```

4. Retrieve unique values using DISTINCT

1. Display unique product categories.

```
SQL> select distinct CATEGORY_NAME FROM Category;

CATEGORY_NAME
--------------------------------------------------
Basic Cases
Designer Cases
Photo Cases
Printed Cases
Protective Cases
Silicone Cases
Transparent Cases

7 rows selected.
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE
  2  FROM PAYMENT;

PAYMENT_MODE
--------------------
Credit Card
Cash
UPI
Debit Card
```

5. Search products based on conditions

1. Find products within a specific price range (₹20,000 to ₹40,000).

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 20000 AND 40000;

no rows selected
```

Phone cases do not have price range of 20,000 to 40,000

So, 500 to 600

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 500 AND 600;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
P000005
Luxury Designer Case
       599

P000020
Elegant Designer Case
       549

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
```

```
P000027
Rugged Protection Case
       549

P000028
Military Grade Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------
     PRICE
----------
       599
```

2. Display products belonging to a particular category.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 'CAT0005';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------
CATEGOR      PRICE
------- ----------
P000005
Luxury Designer Case
CAT0005        599

P000020
Elegant Designer Case
CAT0005        549
```

```
PRODUCT
-------
PRODUCT_NAME
-----------------------------------
CATEGOR      PRICE
------- ----------

P000021
Premium Pattern Case
CAT0005        649

P000022
Luxury Gold Case

PRODUCT
-------
PRODUCT_NAME
-----------------------------------
CATEGOR      PRICE
------- ----------
CAT0005        699
```

3. Find products currently available in inventory.

```
SQL> SELECT * FROM Inventory WHERE Stock_Status = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0001 P000001 S000001             40 Available   12-SEP-26
Inv0002 P000002 S000002             35 Available   13-SEP-26
Inv0003 P000003 S000003             30 Available   15-SEP-26
Inv0004 P000004 S000004             25 Available   14-SEP-26
Inv0005 P000005 S000005             35 Available   16-SEP-26
Inv0006 P000006 S000006             20 Available   17-SEP-26

6 rows selected.
```

4. Search products using product name.

```
SQL> SELECT * FROM Product WHERE Product_Name = 'Luxury Gold Case';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P000022
Luxury Gold Case
CAT0005        699             12
```

5. Find low-stock products

```
SQL> SELECT * FROM Inventory WHERE Stock_Quantity < 30;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0004 P000004 S000004             25 Available   14-SEP-26
Inv0006 P000006 S000006             20 Available   17-SEP-26
Inv0007 P000007 S000001              0 Unavailable 17-SEP-26
```

6. Retrieve customer and product information

1. Display customer details with their orders.

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         O.ORDER_ID, O.ORDER_DATE,
  3         O.TOTAL_AMOUNT, O.ORDER_STATUS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  ORDER BY O.ORDER_ID;

CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C001  Arun Kumar                            1001 20-SEP-26         1299
Pending

C002  Aakash                                1002 21-SEP-26          899
Shipped

C003  Rahul Raj                             1003 22-SEP-26         1599
Delivered


CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C001  Arun Kumar                            1004 23-SEP-26          699
Pending
```

```
CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C002  Aakash                                1013 28-SEP-26          499
Delivered

C003  Rahul Raj                             1014 29-SEP-26          199
Delivered

C004  Divya Kumar                           1015 30-SEP-26          229
Delivered


CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C005  Karthik S                             1016 01-OCT-26          249
Delivered

C001  Arun Kumar                            1017 27-SEP-26          449
Delivered


17 rows selected.
```

2. Display product details with category information.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME,
  2         P.PRICE, P.STOCK_QUANTITY,
  3         C.CATEGORY_NAME
  4  FROM PRODUCT P
  5  JOIN CATEGORY C
  6  ON P.CATEGORY_ID = C.CATEGORY_ID
  7  ORDER BY P.PRODUCT_ID;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000001
Clear Shock Case
       299             50
Basic Cases
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000002
Soft Silicone Case
       249             45
Silicone Cases


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000003
Crystal Clear Case
       199             60
Transparent Cases


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000004
Floral Printed Case
       349             35
Printed Cases
```

```
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000028
Military Grade Case
       599              15
Protective Cases


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000029
Everyday Basic Case
       219              30
Basic Cases


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------
P000030
Simple White Case
       239              25
Basic Cases


30 rows selected.
```

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      C.EMAIL,
  5      C.PHONE
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  JOIN ORDER_DETAILS OD
 10      ON O.ORDER_ID = OD.ORDER_ID
 11  WHERE OD.PRODUCT_ID = 'P000007';

CUSTO CUSTOMER_NAME                    EMAIL                          PHONE
----- ------------------------------ ------------------------------ ----------
C002  Aakash                         aakash35@gmail.com             9876543211
```

4. Display products purchased by each customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         P.PRODUCT_NAME,
  5         OD.QUANTITY,
  6         OD.PRICE
  7  FROM CUSTOMER C, ORDERS O, ORDER_DETAILS OD, PRODUCT P
  8  WHERE C.CUSTOMER_ID = O.CUSTOMER_ID
  9  AND O.ORDER_ID = OD.ORDER_ID
 10  AND OD.PRODUCT_ID = P.PRODUCT_ID
 11  ORDER BY C.CUSTOMER_NAME;

CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
C002  Aakash                               1013
Heavy Duty Case
         1        499

C002  Aakash                               1007
Crystal Clear Case
         1       1200

CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

C002  Aakash                               1002
Crystal Clear Case
         1        899

C001  Arun Kumar                           1004
Luxury Designer Case
```

```
CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1        699

C001  Arun Kumar                           1001
Clear Shock Case
         2        499

C001  Arun Kumar                           1001

CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Soft Silicone Case
         1        301

C001  Arun Kumar                           1017
Personal Photo Case
         1        449


CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
C004  Divya Kumar                          1005
Clear Shock Case
         1        450

C004  Divya Kumar                          1011
Luxury Gold Case
```

```
CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
C005  Karthik S                           1009
Luxury Designer Case
         1        999

C003  Rahul Raj                           1008
Floral Printed Case
         2       1000

CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

C003  Rahul Raj                           1014
Matte Basic Case
         1        199

C003  Rahul Raj                           1003
Floral Printed Case

CUSTO CUSTOMER_NAME                   ORDER_ID
----- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         2      799.5


18 rows selected.
```

7. Apply multiple filtering conditions

1. Find phone case products costing more than ₹600.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, P.PRICE
  2  FROM PRODUCT P
  3  JOIN CATEGORY C
  4  ON P.CATEGORY_ID = C.CATEGORY_ID
  5  WHERE P.PRICE > 600
  6    ORDER BY  P.PRICE DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
P000022
Luxury Gold Case
       699

P000021
Premium Pattern Case
       649
```

2. Search products containing the word “CLEAR”.

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE UPPER(PRODUCT_NAME) LIKE '%CLEAR%';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
P000001
Clear Shock Case
       299

P000003
Crystal Clear Case
       199

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------

P000014
Clear Glitter Case
       249

P000016
Ultra Clear Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
       229
```

3. Find orders placed within a specific date range.

```
SQL> SELECT ORDER_ID, CUSTOMER_ID, ORDER_DATE,
  2         TOTAL_AMOUNT, ORDER_STATUS
  3  FROM ORDERS
  4  WHERE ORDER_DATE >= DATE '2026-09-20'
  5    AND ORDER_DATE < DATE '2026-10-01'
  6  ORDER BY ORDER_DATE;

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ ----------------
      1001 C001  20-SEP-26         1299 Pending
      1002 C002  21-SEP-26          899 Shipped
      1003 C003  22-SEP-26         1599 Delivered
      1004 C001  23-SEP-26          699 Pending
      1010 C004  24-SEP-26          550 Shipped
      1006 C005  24-SEP-26          750 Pending
      1011 C004  24-SEP-26          500 Delivered
      1012 C005  24-SEP-26          550 Pending
      1005 C004  24-SEP-26          450 Shipped
      1007 C002  25-SEP-26         1200 Delivered
      1008 C003  25-SEP-26          550 Shipped

  ORDER_ID CUSTO ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----- --------- ------------ ----------------
      1009 C005  26-SEP-26          999 Pending
      1017 C001  27-SEP-26          449 Delivered
      1013 C002  28-SEP-26          499 Delivered
      1014 C003  29-SEP-26          199 Delivered
      1015 C004  30-SEP-26          229 Delivered

16 rows selected.
```

8. Generate basic business reports

Report 1: Product Availability Report

Display product name, price, stock quantity, and availability status.

```
SQL> SELECT PRODUCT_NAME, PRICE, STOCK_QUANTITY,
  2         CASE
  3             WHEN STOCK_QUANTITY > 0 THEN 'Available'
  4             ELSE 'Out of Stock'
  5         END AS AVAILABILITY_STATUS
  6  FROM PRODUCT
  7  ORDER BY PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Abstract Art Case
       379             25 Available

Blue Silicone Case
       279             45 Available

Cartoon Print Case
       329             35 Available


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Classic Black Case
       229             35 Available

Clear Glitter Case
       249             50 Available

Clear Shock Case
       299             50 Available
```

```
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Collage Photo Case
       499             20 Available

Crystal Clear Case
       199             60 Available

Crystal Edge Case
       269             45 Available


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Custom Memory Case
       479             30 Available

Elegant Designer Case
       549             20 Available

Everyday Basic Case
       219             30 Available


PRODUCT_NAME
--------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Family Photo Case
       449             25 Available

Floral Print Case
       349             30 Available

Floral Printed Case
       349             35 Available
```

```
PRODUCT_NAME
--------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Heavy Duty Case
       499             30 Available

Luxury Designer Case
       599             25 Available

Luxury Gold Case
       699             12 Available


PRODUCT_NAME
--------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Matte Basic Case
       199             40 Available

Military Grade Case
       599             15 Available

Personal Photo Case
       449             40 Available


PRODUCT_NAME
--------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Pink Silicone Case
       279             40 Available

Premium Pattern Case
       649             15 Available

Rugged Protection Case
       549             20 Available
```

```
PRODUCT_NAME
--------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Shockproof Armor Case
       499             25 Available

Simple White Case
       239             25 Available

Slim Basic Case
       249             30 Available


PRODUCT_NAME
--------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Soft Grip Case
       299             35 Available

Soft Silicone Case
       249             45 Available

Ultra Clear Case
       229             55 Available


30 rows selected.
```

Report 2: Customer Report

a. Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
              6
```

b. New customer registrations

```
SQL> SELECT CUSTOMER_ID, CUSTOMER_NAME, REGISTRATION_DATE
  2  FROM CUSTOMER
  3  WHERE REGISTRATION_DATE >= SYSDATE - 30
  4  ORDER BY REGISTRATION_DATE DESC;

CUSTO CUSTOMER_NAME                  REGISTRAT
----- ------------------------------ ---------
C006  Priya Sharma                   10-OCT-26
```

Report 3: Order Report

a. Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          17
```

b. Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'DELIVERED';

COMPLETED_ORDERS
----------------
               8
```

C. Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'PENDING';

PENDING_ORDERS
--------------
             5
```

d. Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDERS
  3  WHERE UPPER(ORDER_STATUS) = 'CANCELLED';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

a. Highest-priced products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE = (SELECT MAX(PRICE) FROM PRODUCT);

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
P000022
Luxury Gold Case
       699
```

b. Most-reviewed products

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME,
  2         COUNT(R.REVIEW_ID) AS TOTAL_REVIEWS
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCT_ID
  6  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVIEWS DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_REVIEWS
-------------
P000005
Luxury Designer Case
           2

P000003
Crystal Clear Case
           2

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_REVIEWS
-------------

P000001
Clear Shock Case
           2

P000002
Soft Silicone Case
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_REVIEWS
-------------
Slim Basic Case
           1

P000008
Matte Basic Case
           1


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_REVIEWS
-------------
P000029
Everyday Basic Case
           1

P000017
Floral Print Case
           1

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_REVIEWS
-------------


30 rows selected.
```

c. Available products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY
  2  FROM PRODUCT
  3  WHERE STOCK_QUANTITY > 0
  4  ORDER BY PRODUCT_NAME;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000019
Abstract Art Case
       379             25

P000011
Blue Silicone Case
       279             45

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P000018
Cartoon Print Case
       329             35

P000009
Classic Black Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       229             35
```

```
Clear Glitter Case
       249             50

P000001

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Clear Shock Case
       299             50

P000024
Collage Photo Case
       499             20


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000003
Crystal Clear Case
       199             60

P000015
Crystal Edge Case
       269             45
```

```
P000025
Custom Memory Case
       479             30

P000020
Elegant Designer Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       549             20

P000029
Everyday Basic Case
       219             30

P000023

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Family Photo Case
       449             25

P000017
Floral Print Case
       349             30
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000004
Floral Printed Case
       349             35

P000007
Heavy Duty Case
       499             30

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P000005
Luxury Designer Case
       599             25

P000022
Luxury Gold Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       699             12

P000008
Matte Basic Case
       199             40
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       699             12

P000008
Matte Basic Case
       199             40

P000028

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Military Grade Case
       599             15

P000006
Personal Photo Case
       449             40


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000012
Pink Silicone Case
       279             40

P000021
Premium Pattern Case
       649             15
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P000027
Rugged Protection Case
       549             20

P000026
Shockproof Armor Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       499             25

P000030
Simple White Case
       239             25

P000010

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Slim Basic Case
       249             30

P000013
Soft Grip Case
       299             35
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000002
Soft Silicone Case
       249             45

P000016
Ultra Clear Case
       229             55

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------


30 rows selected.
```